

In-Vehicle Transitional Interfaces: User Journey in Virtual Reality

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Figure 1: Augmented reality windshield display interface customized by a user in virtual reality.

ABSTRACT

In this poster, we investigate novel user-centered methods and techniques for cross-virtuality user journeys for designing in-vehicle experiences. We argue that rapid prototyping of automotive human-machine interfaces (HMIs) can be achieved with mixed reality technology along the reality-virtuality (RV) continuum. For example, standalone virtual reality (VR) head-mounted displays (HMDs) coupled with hand-worn controllers can be utilized to prototype AR interfaces such as windshield displays (WSDs) in order to facilitate an increased level of visual and spatial perception. Subsequently, the prototyped design in VR could be transferred to an AR interface (e.g., using an AR HMD) for further refinements in real-world

settings. We believe that transitional interfaces in the automotive domain have the potential to fundamentally improve the design and development of novel in-vehicle visualization and interaction modalities, as well as to enhance user-centered and non-driving related collaborative activities for automated vehicles.

CCS CONCEPTS

• **Human-centered computing** → **Mixed / augmented reality**; **Virtual reality**; **Interaction techniques**; **User studies**; **Scenario-based design**; **Interface design prototyping**.

KEYWORDS

mixed reality, intelligent vehicles, automated driving, transitional interfaces

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1 INTRODUCTION

Augmented Reality (AR) windshield displays (WSDs) provide a large 3D space for displaying content, while the driver can keep an overview of the outside road environment [14, 17]. Therefore, the windshield display is a novel in-vehicle canvas for presenting and interacting with content in automated vehicles (AVs) [1, 18]. On this canvas, we plan to ideate and prototype new user experiences based on vehicle safety, and usability of performing non-driving related tasks (NDRTs). Several automakers have already applied AR head-up displays to recent models (e.g., Audi E-tron, Kia EV6, among others), focusing on the primary driving tasks, such as steering and accelerating. We are now in a transition period of driving in mixed traffic with both human-operated vehicles, and (conditionally) automated vehicles [16]. Therefore, not only driving information, but also NDRTs (e.g., entertainment and work-related content) for user convenience and enjoyment may be performed on AR WSDs in the future [5]. Recently, design studies for the tertiary driving task related content have investigated which design elements designers should consider for the AR WSD [13]. Studies on design layout, content, color, transparency, readability, and size of the content provide the guidelines for future WSD design [2, 9, 14, 15].

It is expected that advancements in both AR and VR technology will enter vehicles as well [10–12]. While AR HUDs are already being introduced, in-vehicle VR still needs to address some challenges [3]. Furthermore, combining AR and VR, and traditional displays (e.g., dashboards, smartphone/tablet screens) into transitional interfaces presents even more challenges, but also opportunities [7]. In our view, cross-virtuality [8] mechanisms allow a seamless integration and transition between traditional 2D displays, and 3D AR windshield displays, in order to provide automotive users (i.e., drivers, passengers) with immersive visual and interaction support, depending on their current non-driving related tasks, such as work and entertainment activities. However, the comprehensive and holistic design direction that can examine the user preferences for transitional interfaces has not been explored yet.

Therefore, in our present work-in-progress, we aim to develop a prototyping tool that allows users to design and arrange AR contents directly on the WSD in our VR simulator (see Figure 1). Furthermore, we plan to observe and analyze the preferences and user journeys of the participants. Finally, we propose a cross-reality transitional interface along the reality-virtuality (RV) continuum [4] without interruptions of the seamless mixed reality user experience.

2 DESIGN AND IMPLEMENTATION

We do not expect participants to be technologically-proficient, or to have prior experiences with VR, thus, the user journey process consists of three phases including a tutorial and a user-friendly familiarization task to have participants ease into the VR user journey.

The VR simulator is developed in Unity, and composed of the following parts: WSD as a canvas, workspace where users can select contents including drawing tools, and the different content

windows for performing non-driving related tasks (NDRTs). The main environment of the VR space is the vehicle interior. Participants can put on the VR HMD (HTC Vive Focus 3), and utilize VR hand-worn controllers or pens to design their preferred user interface (UI) on the WSD while sitting in the driver's seat. As one of the ideation metaphors, participants put post-it notes on the whiteboard or draw with a pen to resemble natural physical experiences in the VR simulator. The following interaction types can be performed by the participants in the VR workspace:

- Create and remove content windows on the WSD
- Grab and move content windows to the any position on the WSD
- Adjust the size of content windows
- Adjust the opacity/transparency of the content window
- Select the background color for content windows in the menu panel
- Draw and erase the self-made drawings/notes on the WSD
- Clear all contents of the WSD in order to restart with an “empty” WSD
- Take a “snapshot” of the current WSD configuration for future use in order to pick up where you left off

VR Familiarization. When the VR simulator begins, participants are presented with a tutorial video containing information about the purpose of each ideation task, and how to utilize the VR controllers in the drawing workspace. The environment of the tutorial scene is a living room which resembles a familiar and comfortable setting while sitting on a sofa watching the video on a virtual TV (see Figure 2).

Windshield Display Personalization. In this phase of the user journey, we provide the practice session, in which the participants can perform the task just as they learned from the tutorial video in the familiarization phase. The following steps are arranged to try the single tasks per step. Participants are encouraged to place, move, adjust, and draw content windows and text content in any way they like on the windshield display inside the vehicle. As a visual metaphor, we present the content windows as sticky notes that can easily be arrange using the VR controllers.

We provide several types of content windows, according to Pflöging et al. [5] and Riegler et al. [15], that can be placed on the WSD (see Figure 3):

- Vehicle-related information, including current speed and fuel/battery indicators
- Work-related information, including email and workplace messaging content
- Entertainment information, including music and video players
- Social media information, including content from popular social media platforms and chat applications
- Canvas for freely drawing or taking notes using joysticks or a VR pen tool

Each content window has two color options of black and white or colourful. With the right-handed controller, participants can adjust the position, size, and transparency of content windows. The position of the content window is three-dimensional to simulate AR functionality within VR [6]. Therefore, participants can tilt

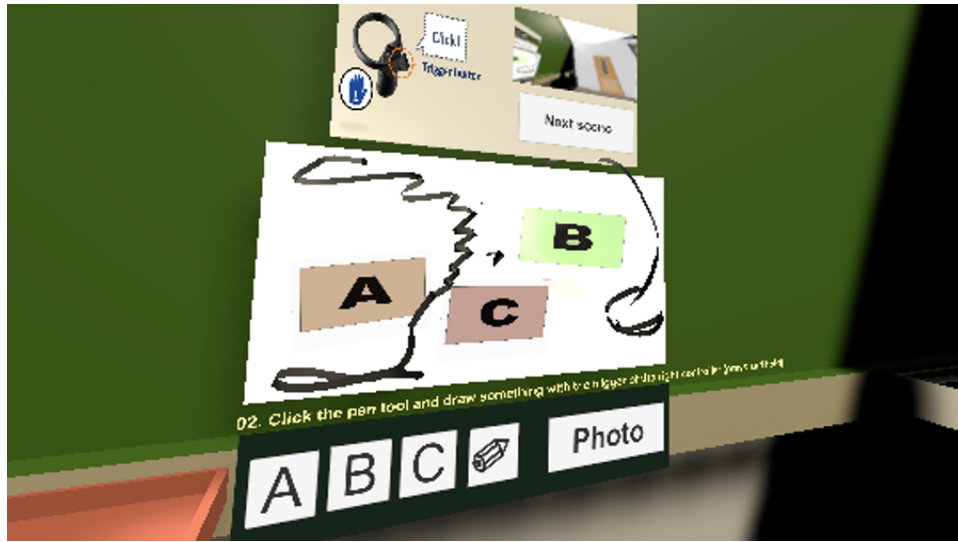


Figure 2: The familiarization task of the user journey. Participants use interaction types provided by the controllers (e.g., drawing, writing, adjusting content windows).



Figure 3: The main interaction task of the user journey. Participants can customize their preferred WSD contents.

content windows depending on their comfortable seating position. The size of content windows can also be modified using the left and right buttons of the thumbstick on the right-handed controller. The transparency of the content window can be fine-tuned by clicking up or down on the thumbstick. In addition for participants being able to design visual properties of the content windows, they can also draw or write freely using the pen tool. Participants can draw a stroke line using the trigger button on the right-hand controller and erase it by pressing the side button. Finally, participants can also arrange the driving information on the WSD according to their preferences.

3 USER STUDY PLANS

While we have tested our prototypical implementation during development with a few participants, we plan to conduct a full user study that validates our cross-virtuality transitional interface. The presented VR user journey represents the initial part of immersive prototyping, and the subsequent AR journey aims to seamlessly transform the content generated in VR into the AR WSD interface.

In the work-in-progress session, we intend to discuss our user study plans with conference participants and aim to find potential collaborations on in-vehicle transitional interfaces.

4 CONCLUSION

In this poster, we presented our research plan and approach to conduct seamless user studies in VR for the purpose of designing user-centered in-vehicle experiences. In particular, we focus on the strengths of each point of the reality-virtuality continuum. For example, in the VR spectrum, we can immersively simulate content in a multitude of virtual vehicle environments and settings, while in the AR spectrum, we can fine-tune content displayed on the WSD in a real vehicle for further validation. The overarching aim of this project is to create seamless transitions on the reality-virtuality continuum, in order to (1) avoid breaking immersion, and (2) support transitional interfaces between AR and VR content displayed on the windshield and head-mounted displays.

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REFERENCES

- [1] Renate Häuslschmid. 2018. Extending head-up displays: exploring the potential of large & 3D automotive windshield displays. (2018).
- [2] Renate Häuslschmid, Donghao Ren, Florian Alt, Andreas Butz, and Tobias Höllerer. 2018. Personalizing content presentation on large 3d head-up displays. *PRESENCE: Virtual and Augmented Reality* 27, 1 (2018), 80–106.
- [3] Mark McGill, Julie Williamson, Alexander Ng, Frank Pollick, and Stephen Brewster. 2020. Challenges in passenger use of mixed reality headsets in cars and other transportation. *Virtual Reality* 24, 4 (2020), 583–603.
- [4] Paul Milgram, Haruo Takemura, Akira Utsumi, and Fumio Kishino. 1995. Augmented reality: A class of displays on the reality-virtuality continuum. In *Telemanipulator and telepresence technologies*, Vol. 2351. International Society for Optics and Photonics, SPIE, 282–292.
- [5] Bastian Pfleging, Maurice Rang, and Nora Broy. 2016. Investigating user needs for non-driving-related activities during automated driving. In *Proceedings of the 15th international conference on mobile and ubiquitous multimedia*. 91–99.
- [6] Eric Ragan, Curtis Wilkes, Doug A Bowman, and Tobias Hollerer. 2009. Simulation of augmented reality systems in purely virtual environments. In *2009 IEEE Virtual Reality Conference*. IEEE, 287–288.
- [7] Andreas Riegler, Christoph Anthes, Clemens Holzmann, Andreas Riener, and Shiva Mohseni. 2021. Autosimar: In-vehicle cross-virtuality transitions between planar displays and 3d augmented reality spaces. In *ISS'21: Interactive Surfaces and Spaces*.
- [8] Andreas Riegler, Christoph Anthes, Hans-Christian Jetter, Christoph Heinzl, Clemens Holzmann, Herbert Jodlbauer, Manuel Brunner, Stefan Auer, Judith Friedl, Bernhard Fröhler, et al. 2020. Cross-Virtuality Visualization, Interaction and Collaboration.. In *XR@ ISS*.
- [9] Andreas Riegler, Andreas Riener, and Clemens Holzmann. 2019. Adaptive dark mode: investigating text and transparency of windshield display content for automated driving. *Mensch und Computer 2019-Workshopband* (2019).
- [10] Andreas Riegler, Andreas Riener, and Clemens Holzmann. 2020. A research agenda for mixed reality in automated vehicles. In *19th International Conference on Mobile and Ubiquitous Multimedia*. 119–131.
- [11] Andreas Riegler, Andreas Riener, and Clemens Holzmann. 2021. A systematic review of augmented reality applications for automated driving: 2009–2020. *PRESENCE: Virtual and Augmented Reality* (2021), 1–80.
- [12] Andreas Riegler, Andreas Riener, and Clemens Holzmann. 2021. A systematic review of virtual reality applications for automated driving: 2009–2020. *Frontiers in human dynamics* (2021), 48.
- [13] Andreas Riegler, Andreas Riener, and Clemens Holzmann. 2022. Content presentation on 3D augmented reality windshield displays in the context of automated driving. In *2022 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*. IEEE, 543–552.
- [14] Andreas Riegler, Klemens Weigl, Andreas Riener, and Clemens Holzmann. 2020. StickyWSD: Investigating content positioning on a windshield display for automated driving. In *19th International Conference on Mobile and Ubiquitous Multimedia*. 143–151.
- [15] Andreas Riegler, Philipp Wintersberger, Andreas Riener, and Clemens Holzmann. 2018. Investigating user preferences for windshield displays in automated vehicles. In *Proceedings of the 7th ACM International Symposium on Pervasive Displays*. 1–7.
- [16] SAE On-Road Automated Vehicle Standards Committee. 2021. Taxonomy and definitions for terms related to on-road motor vehicle automated driving systems.
- [17] Ronald Schroeter and Fabius Steinberger. 2016. Pokémon DRIVE: towards increased situational awareness in semi-automated driving. In *Proceedings of the 28th australian conference on computer-human interaction*. 25–29.
- [18] Gesa Wiegand, Christian Mai, Kai Holländer, and Heinrich Hussmann. 2019. Incarar: A design space towards 3d augmented reality applications in vehicles. In *Proceedings of the 11th international conference on automotive user interfaces and interactive vehicular applications*. 1–13.